



Bokaro Steel Plant / Steel Authority of India Limited

Project Report on

Server Based Alcatel – Lucent PBX

Submitted by:

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INTRODUCTION

Electronics and Telecom Dept. in Bokaro Steel Plant is looking after the following services:

I) TELEPHONE COMMUNICATION

(A) Main Exchanges - Following Main Exchanges are catering to the needs of Telephone Communication in Offices and Residences of Bokaro Steel Plant.

Plant EPAX: Capacity 5000 lines (72xxx, 73xxx, 74xxx and 75xxx and 76xxx series).

T/S EPAX: Capacity 5000 lines expandable up to 7400 lines (84xxx, 85xxx, 86xxx, 87xxx, & 88xxx).

BGH RLU: Capacity 500 lines (89000 to 89499).

Admin. Bldg. EPAX: Capacity 3000 lines (80xxx, 81xxx and 82xxx series).

- ✓ All BSL Telephone Exchanges are having DID facility for Direct Inward Dialing from BSNL Exchange. Hence all the public as well as Mobile Subscribers from anywhere in the world can contact BSL Telephone users.
- ✓ All the residential Telephone Connections have been provided with DOD (Direct Outward Dialing) facility to BSNL phones, within B.S. City.
- ✓ All the BSL exchanges are connected with OFC networks and are working with Unified dialing system

B) Dispatcher/EPABX systems

All the major shops and VIP locations have been provided with Telephone Dispatcher/EPABX system for Internal Communication within the shop. Total systems provided for this purpose – 50 nos., which includes the Dispatcher systems for Telemetry and Railway yard, Communication systems.

C) ED (W) Dispatcher Conference System

A 100 line Ex-Com (EPABX) is provided for Morning and Evening ED (W) Dispatcher conference System for Audio Conference with the HODs of the Plant.

D) Direct Communication Systems

For Point to Point Communication service, there are Direct Communication facilities provided for Hotline Communication between Important Locations.

II) RAILWAY SIGNALLING & COMMUNICATION SYSTEMS

For the purpose of safe and speedy Rail Traffic movements, Railway Signaling & Communication are provided at the following locations, spread throughout the plant. These facilitate Motor Operated Rail Track Control Systems and Yard Voice Communication with Loud Speaking & Talkback Systems for important Railway Yards of the Plant.

- Steel Works Station - With 93 nos. of Motor Operated point machines and Yard Communication System.
- Rolling Mill Station - With 44 nos. of Motor Operated Point machines and Yard Communication system.
- BF Station - With 24 nos. of Motor Operated Point machines and Yard Communication System.
- CCD Station - With 16 nos. of Motor Operated Point machines and Yard Communication System.
- Central Station - With 11 nos. of Motor Operated Point machines and Yard Communication System.
- Post Tripling Yard Station - With Yard Voice Communication System.

- PLC based Automatic Rail Road Level Crossing Audio Visual Systems/ Shop entry Signal Systems covering 19 locations throughout the Plant viz: BF1, BF3, CRM, HRCF, HSM, CCS, Duggal gate, CRM-II and SSD areas.
- Motorized Electro Mechanical Drop Gate systems at Plant Main gates, BF1 and RERS/ CCS (with Data Logging facility) to enhance Safety & Speediness in Road Traffic movements of the Plant.

III) TELEMETRY SYSTEMS

For Centralized Monitoring and Control of different process parameters required for the operations of the Plant - Telemetry/ SCADA Systems have been provided at the following locations.

- Gas System for EMD
- Sintering Plant System for Sinter Plant Process
- Power/SLCC System for DNW, TPP & CPP
- Water System for Water Management Dept.
- Metallurgy System/ EIS (Executive Information System) for Plant Control and MIS purposes of the Plant.

IV) SPECIAL INSTRUMENTS

For Quantitative and Qualitative Analysis of Raw Materials, Intermediate Products & Finished Products - total 13 Analyzers viz: X-Ray Fluorescence Spectrometers, Optical Emission Spectrometers, O₂/ H₂/ N₂ Analyzers etc. are used at various Shop Laboratories. These Systems are operated at Express Laboratories of SMS-I, SMS-II, CCS, BF and Central Lab and which are maintained by Special Instruments Group of the Department.

V) ELECTRONIC CARD REPAIR LAB

As most of the Electronic & Telecom equipments are of sophisticated type, in-house Electronic card repair has become an essential requirement. Most of the Electronic repairs are being carried out in this Repair Lab. This Lab is equipped with QMAX (QT2128) PCB Fault Diagnostic System, System8 PCB Fault locator, Repair / Rework station for SMD Soldering / Disordersing facilities.

Repairing of Wireless, Telephones, Amplifiers, Power Supplies, Chargers, Electronic PCBs etc are being done in this Lab.

VI) OTHER ELECTRONICS AND COMMUNICATION SERVICES

In addition to the above Services/ Systems, the following Communication Systems are also being maintained by Electronics and Telecom(W) dept. are

- Loud Speaking and Talk Back Systems for Voice Communication within the shops like HRCF, CRM, HSM, SMS-II, CCS, CDI, TBS, RMHP etc. total about 400 stations provided throughout the Plant.
- SLIC - Selective Loud speaking Intercom Systems have been provided at CCS, CRM, and HSM etc. for smooth operational activities.
- CCTV Systems - For close monitoring of inaccessible Process Areas of Slabbing Mill, Hot Strip Mill, SMS-II, BF and CCD are provided with Closed Circuit TV system. Total 32 locations are provided with this purpose.
- Video Surveillance Systems:
 - ✓ At 6 nos. of Plant Gates, for monitoring of movement of men and materials, around the periphery of the Plant.
 - ✓ For monitoring of incoming rakes of coal at Weigh Bridge # 3, 4 & 5

- ✓ For monitoring the movement of vehicles at OG, MES-1, MES-2, and RERS (CCS).
- ✓ For monitoring the movement of man & material in Slag Dump area.
- PA system / General Broadcasting Systems are being provided for day to day activities in Plant as well as in Township.
- Fixed Wireless Communication Systems- These have been provided for operational usage of Shops viz: RMHP, SGP, SMS2, HSM, CRM and Traffic.
- Handheld Wireless Set (VHF) - There are around 720 nos. of Handheld Wireless Sets being used for Wireless Communication purpose throughout Plant.
- Power Line Carrier Communication (PLCC) System - PLCC systems have been provided for Signaling, Protection and Voice Communication facilities over 132 KV HT lines between BSL and DVC/ CTPS - for Power Supply requirements of the Plant.

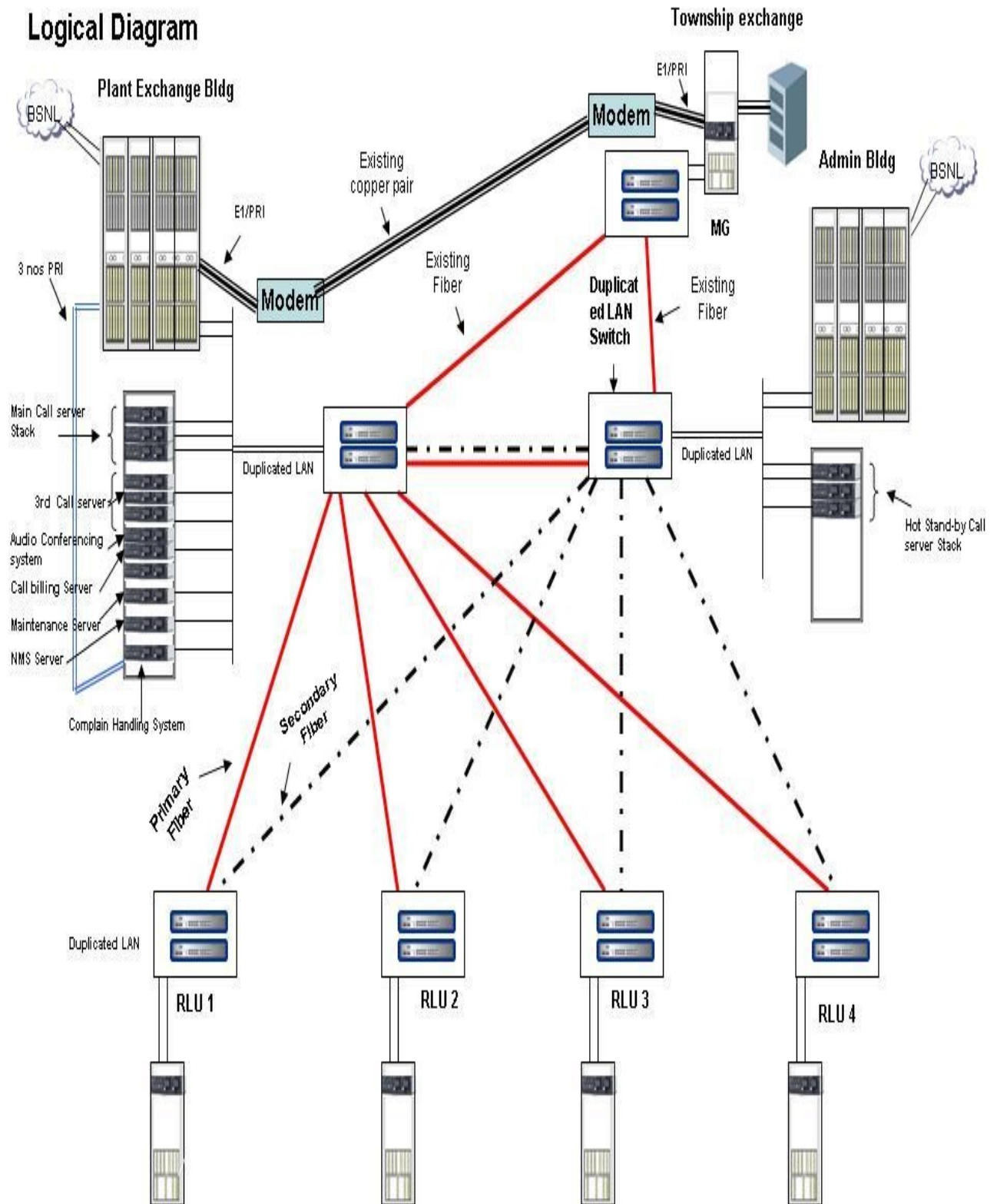
Plant Exchange:

The recently commissioned new server based exchange was commissioned in May, 2013. The proposed voice network consists of two independent PBX locations at Plant and Admin building wired for 7000 and 3000 subscribers respectively. For ensuring geographical redundancy 5 numbers of passive servers also installed at Coke Oven, Blast Furnace, CRM 1 & 3 and Town ship locations which are wired for 500 connections respectively. The proposed system architecture consists of media gateways at individual locations and call servers installed at Plant and ADM locations. The servers installed at locations Plant and ADM also runs on Hot-Stand by to ensure the real time back up of system.

The network of media gateways is connected with data network installed through various Hubs and switches at different RLU and central locations. Additional facilities like Audio Conference Bridge, Voice mail system and IVRS based complaint handling system are also part of the package. The exchange further wired for 2000 extensions provided the hardware is enabled.

LOGICAL DIAGRAM

Logical Diagram



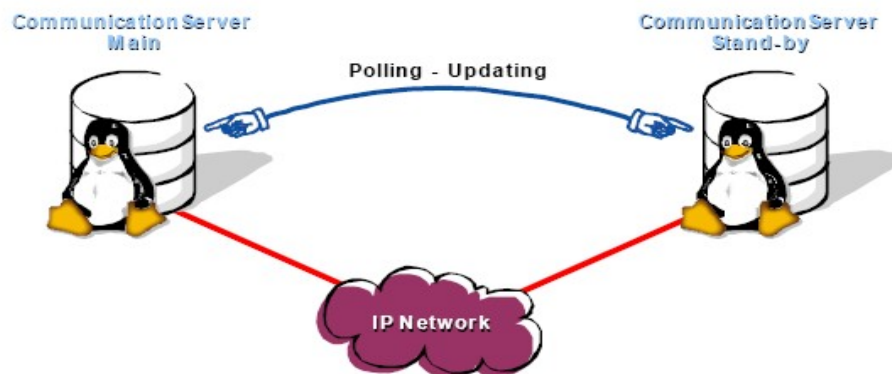
Call-Server functionality:

The proposed Server based IP telephony solution consists of two main sites Viz., Plant Exchange bldg. & Admin bldg. Another category of sites comprising of 04nos small remote sites called Remote Shops and one no. Township exchange. All the sites are connected over IP through campus wide data network. Main Call Server stack consists of three HP servers which will be installed at Plant Exchange bldg. Another stack of servers consists of three HP servers will be installed at Admin bldg. In ideal scenario call processing will be handled by the main server stack installed at Plant Exchange. In case of main call-servers failed the call processing will be taken over by the hot stand-by call server stack installed at Admin bldg. One more stack of Call Servers (three no's of HP servers) has been provisioned at Plant Exchange will work at Cold standby Call Servers. In case of Active & hot standby serves fail these will be manually plugged into the network and take over the call processing.

Call-sever redundancy & fail over functionality:

In this type of configuration, two communication servers coexist in the same system. One server is active, and is the primary communication server (Call server stack at Plant Exchange). The other server is constantly in watchdog mode on standby (call server stack at Admin bldg.). If the primary server fails, the standby automatically takes over. The standby communication server is updated continuously and is ready to act as the primary server at any time. The power of the Alcatel-Lucent OmniPCX Enterprise Communication Server redundancy mechanism is that ALL data, including databases, applications, and communication-handling software, is run in parallel on both servers. This ensures a reliable, secure switchover from one server to the other. During normal operation, a polling dialog is established and maintained

between the main and standby communication servers. Interruption of this dialog indicates to the standby machine that the main communication server not available. The standby communication server then takes over as the main communication server.



Media Gateways at Plant Exchange:

Another main component of this solution is ACT28 and ACT14 Media Gateways. ACT28 provides 28nos of slots and ACT14 provides 14nos of slots to house the different resource cards like eUA32, eZ32, INT-OF, INT-IP etc.

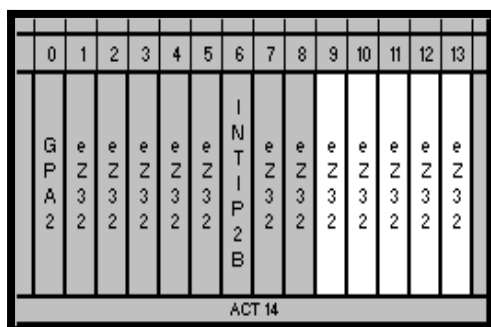


Figure: ACT14

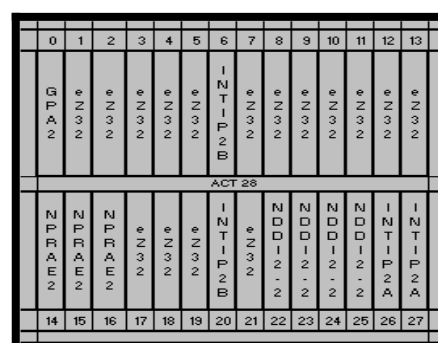


Figure ACT28

Functions of Peripheral cards as below:

1. ACT14 : Media gateway with 14nos of slots.
2. ACT28 : Media gateway with 28nos of slots.

3. eZ32 : 32 ports Analog interface card
4. eUA32 : 32 ports Digital interface card
5. NDDI2-2 : 8 port Analog trunk interface card
6. NPRAE-2 : 2port 2mb PRI interface card
7. INTOF2 : peripheral card used to interface two ACTs
8. INT-IP3 : IP interface card – connects ACT with the Ethernet switch
& its DSP daughter boards provides VoIP resources
9. GPA2 : Voice Guides and conference resource card
- 10.4635H Voice Mail : Voice mail service

Media Gateways at Admin Building:

As explained earlier the stack of hot stand-by server (3nos of servers) will be installed here. Seven no's of ACT28 will be installed for housing required resources cards.

Media Gateways with Local survivability:

One ACT-14 will be installed at each remote shops to house required resource cards. This Media Gateway is IP enabled and get registered with the main call-server over campus LAN. Same way one ACT-28 will be installed at Township.

In case of outage of LAN network with Plant Exchange & Admin building Call Servers OR failure of both the Call server stacks, the Media Gateways will survive with the help of one 3rd Local call server called Passive Communication server (PCS). The Passive Communication Server (PCS) is a 3rd server to provide the continuity of telephony services to the Telephony users in case of:

1. Loss of the OXE's Call Servers (CS), duplicated or not (e.g. massive outage in the central site.
2. Breakdown of the IP links between the remote sites and the central site.

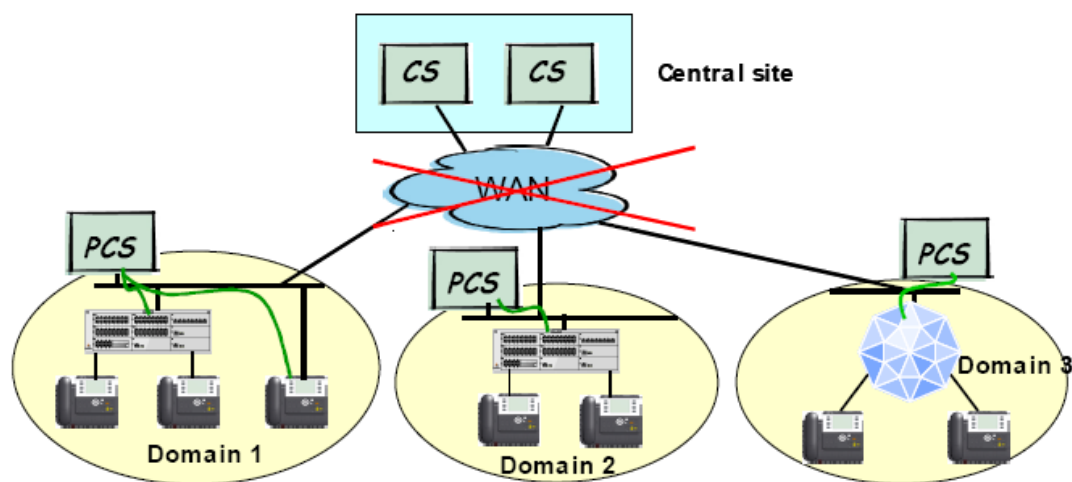
Duplicated CS Database updating:

The PCS database is an exact copy of the CS database.

Two update modes are available:

1. Manually by system command on the main CS (pcscopy)
2. Automatically at a predefined configurable time
 - Daily (hour)
 - Or weekly (hour, day)

A typical example of the PCS deployment is shown in the figure below:



A pooling (keep alive based) is used between the main CS and the PCS thru a signaling IP link to know the IP network status between the Servers. A PCS becomes Active in case of rupture of the IP link with the CSs (A PCS can secure an equipment only if it is in Active status). When the equipments of an

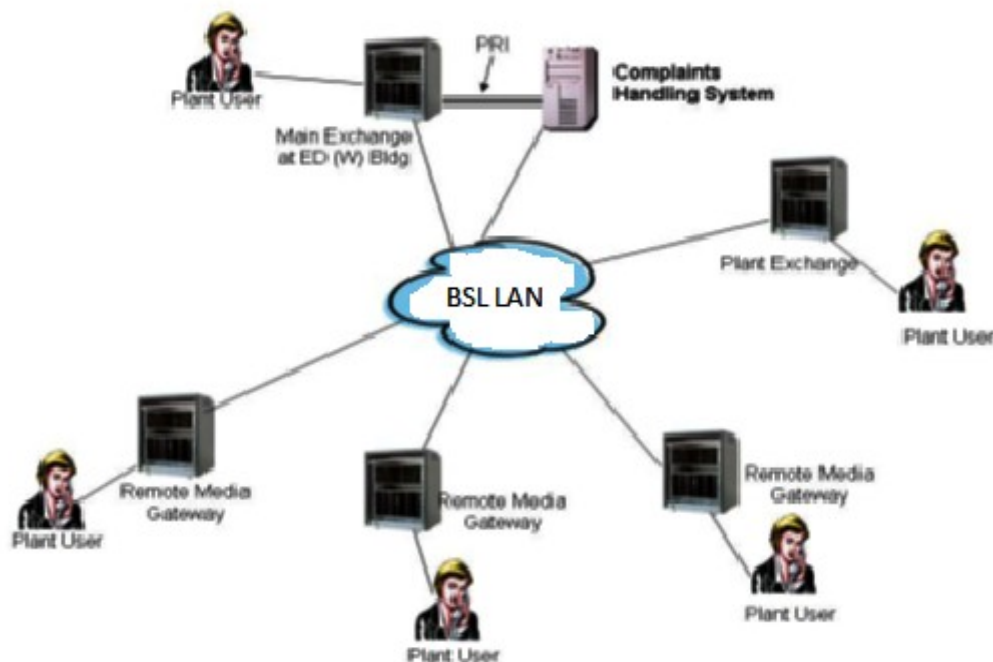
IP domain (IP Touch and Media Gateways) lose the IP link with the CS, they try to establish an IP signaling link with their PCS, which is Active, and finally become attached to it. Then, The PCS and the equipments attached to it have to be considered as a Stand-Alone System. All the media gateways and IP Phones take a reset before registering to the PCS server. The typical downtime is approximately 5 min. before the IP phones and Media Gateways start working.

When PCS is in active state, it provides the basic telephony feature to the rescued domain. The IP Phones will work as a normal IP Extension and will allow the agents to make outgoing calls manually. When the PCS detects that the IP link with the CS(s) is back, it launches a timer. When this timer expires, all the devices attached to the PCS are reset and becomes attached to the CS.

The timer configuration allows 3 reset modes:

- by default: the reset is processed 30 s after IP link reestablishment
- Hour: a specific hour (HH:MM) is configured. The reset is processed at this hour.
- Timer value: Duration in seconds (from 1 to 65535) is configured. The reset is processed at the end of this timer: If the timer = 0, the PCS doesn't reset the IP equipments (reset has to be done manually).

Complaint handling system:



The proposed system answers an incoming call through an automated fault logging system. This will be installed at Plant Exchange. Caller may ask to either book a complaint or find the status of a complaint. If the Caller books a new complaint, he or she needs to punch in the faulty extension number or telephone number. The application can ask the caller for alternate contact number. The system picks up the status of the booked number accordingly system prompts for booking the complaint or reject the punched in number. The booked complaint will be deposited in the complaint handling section who is handling the complaints of that particular equipment. The sections are provided with client terminals where in the technician can get the complaints printed on his terminal. The designated supervisor then input the necessary fault codes and updates the complaint status. Supervisor can get the complete history of a complaint at any point of time through the terminal. The application should be divided in to three main modules.

Main Unit: An IVRS system will accept an incoming call and take the complaints from the callers complaints. This IVRS will be connected with the Server based exchange through 3 nos of PRI trunk. It also facilitates the technician to update the status of a complaint.

Terminal Utilities: A supervisor can view all the pending complaints and can update them. He can also build up the database for his department. Option to print the docket numbers online on the printer as and w hen a complaint is booked.

Voice Mail System – 4635H

The Alcatel-Lucent 4635h Voice Messaging System is integrated into the OmniPCX Enterprise for more efficient management of the greeting functions. The Alcatel 4635 is a voice mail system which is integrated in a shelf in ACT28 installed at Plant Exchange. The Alcatel 4635 messaging system is used to exchange, with a correspondent (person, mailbox), information in the form of voice messages or faxes, store these messages or faxes and restore them on demand to the recipient. The Alcatel 4635 software is supplied in a standard version (voice mail) to which different options can be added. These options include:

- specific mailboxes

The basic hardware configuration comprises:

- 1 VPM35 (Voice Processing Multiple35) board, for connecting an IDE hard disk.

- Optionally, 1 or 2 MSBI (Mass Storage Board IDE) with one or two hard disks each (1.4GB or 2.16GB capacity). They contain the software, voice guides, database and messages.
- 1 to 8 SPA3 (Speech Processing Auxiliary) boards. They each offer 8 channels with simultaneous access to the messaging system. Optionally, 1 ATB2 board (or ATB31 where fitted with 2 MSBI boards). Backplane complement, for the IDE bus interconnection between VPM35 and MSBIs.
- 1 V24 cable connecting the VPM35 board to a VT100 type console, called SMT (System Manager Terminal), which must remain dedicated to the Alcatel 4635H messaging system. You can use the Alcatel 4400 PBX system console to emulate the VT100 terminal. The cable can also be connected to the switch box, for remote maintenance. A printer connected to the console is used to output reports and statistics.
- 1 special Ethernet cable for connecting the VPM35 board to the customer LAN.

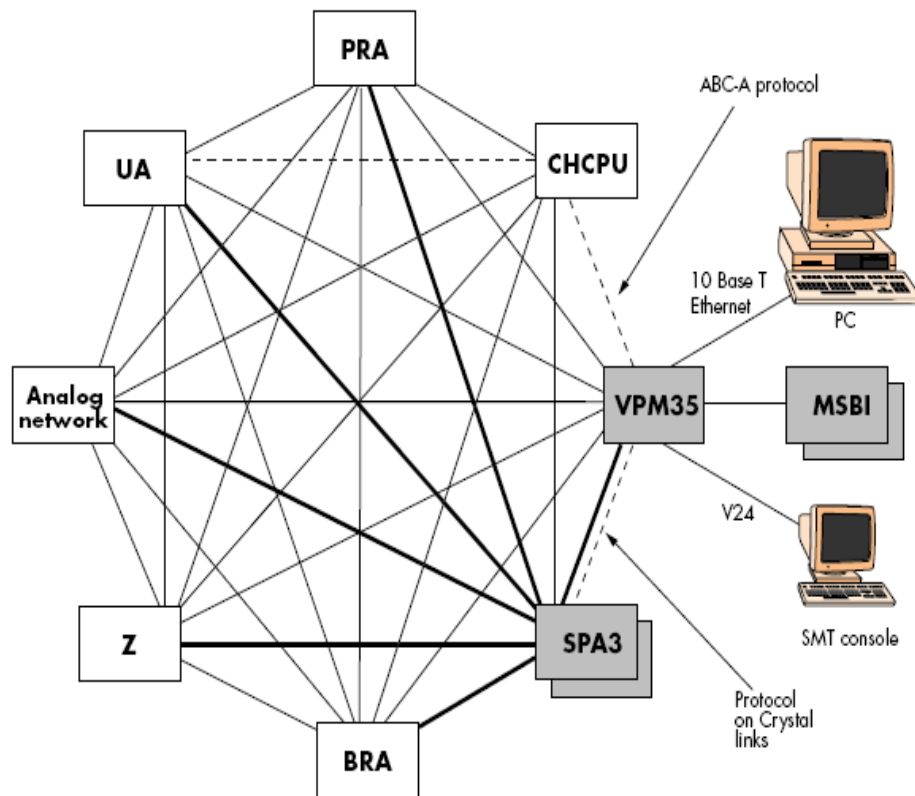


Figure: Hardware architecture(Alcatel 4635H (R.2) version)

Audio Conference Bridge - Omni Touch Team Work Server:

Omni Touch My Teamwork is an application enabling employees to collaborate remotely with colleagues or external participants. With Omni Touch My Teamwork:

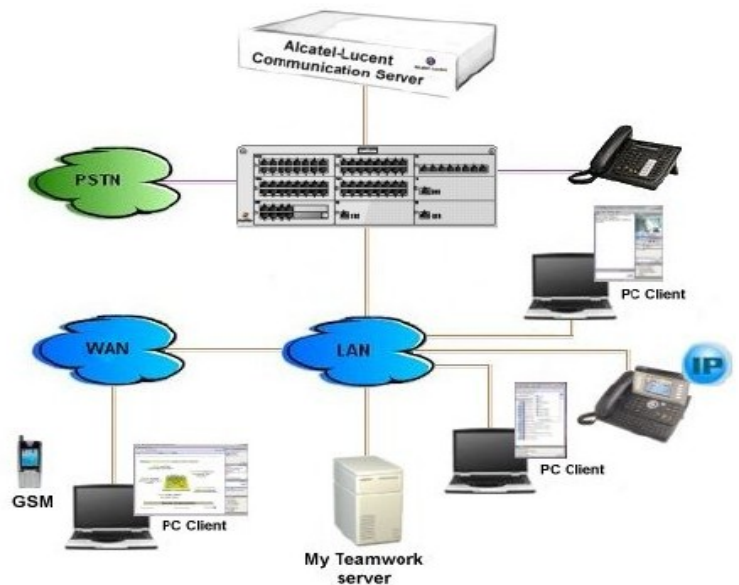
- Team productivity is increased: internally and with your external contacts.
- You save time and money: no need to be physically in the same room to work on the same document (thus saving on travel expenses).

- c. You collaborate easily: easy to deploy and use (even remotely); all it requires is a Web browser (no software installation).

Architecture

Using a customer/server architecture and integrating SIP and XML technologies, you can access OmniTouch My Teamwork from your IP Touch 4068 or 4038 sets, or even from a PC.

What's more, because OmniTouch My Teamwork is browser based, your employees can access it when they are out of the office.



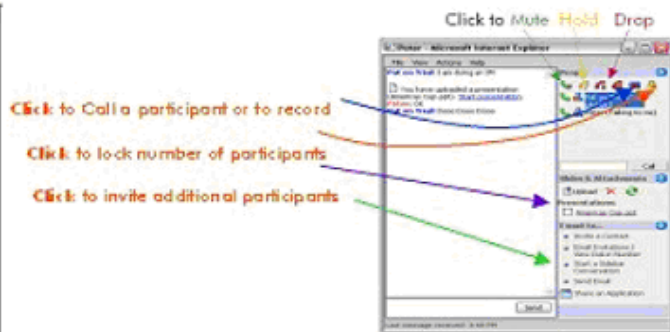
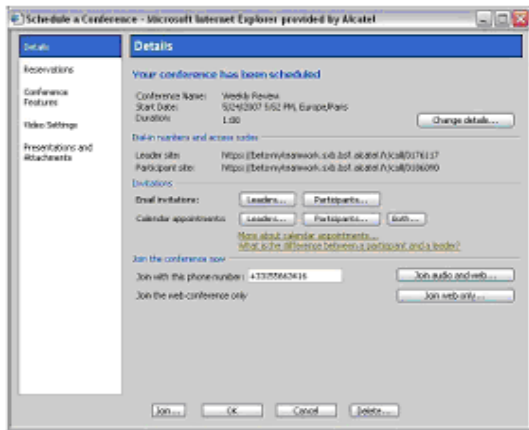
Overview

A graphical, intuitive and friendly interface allows users to easily organize and manage their conferences.

Audio conferencing

You can improvise audio conferences in a few clicks on the homepage or else program the details in advance: determine the length, access type, send invitations to your contacts (automatic notification by e-mail etc.) callers (depending on number of audio ports available) can participate in your casual or programmed conferences.

Meet - me conference Casual conference

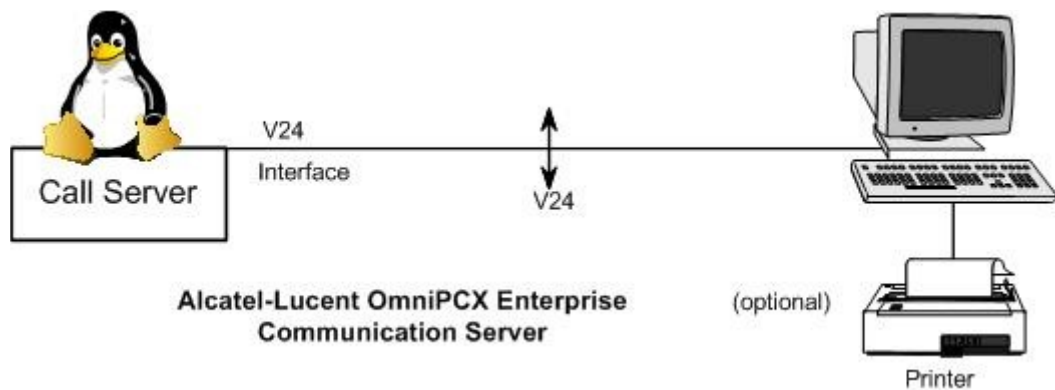


General maintenance Commands

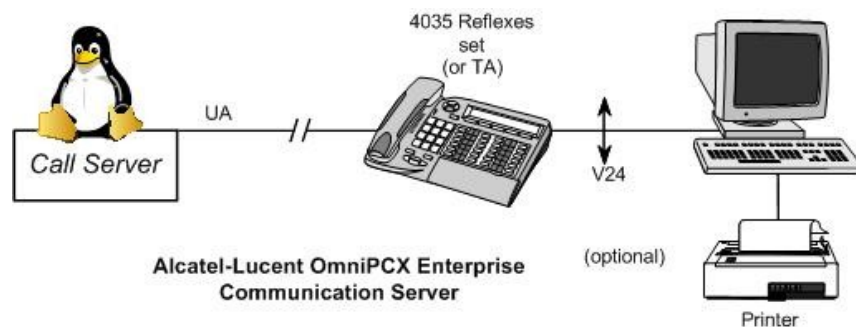
Connection

The VT100/Televideo terminal¹ can be connected in different ways:

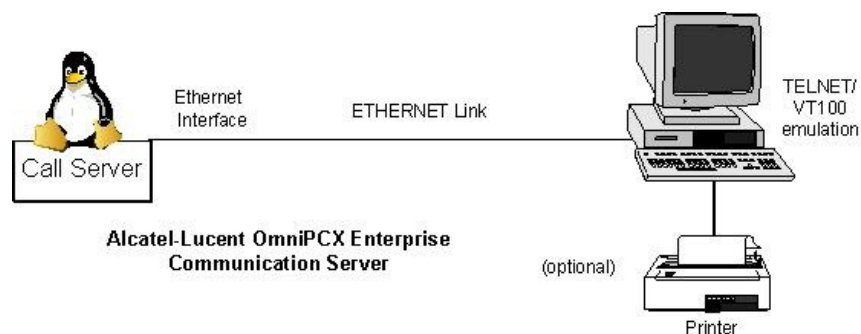
- V24 connection



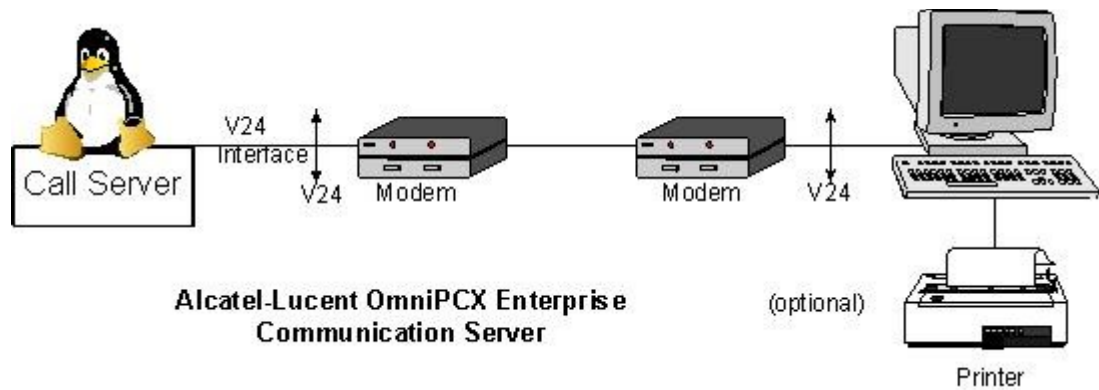
- Connection via UA equipment



- Ethernet® connection



- V24 connection via modem



Use of Commands:

Before using any of the commands be sure of its result. Some commands may disrupt the entire service of the communication network.

Logon to the Communication Server:

Welcome to pcx2a

Alcatel-Lucent OmniPCX Enterprise

login: <Login ID>

Password:<Password>

Last login: Mon Sep 17 11:34:41 from 10.143.71.60

Alcatel-Lucent OmniPCX Enterprise

standard installation last performed: 20-Jan-2012 10:42:42

The role of the CPU is MAIN

Application software identity

R9.1-i1.605-37-a-ln-c0

Business identification: R9.1

Release:

DELIVERY i1.605

Patch identification: 37

Dynamic patch identification: a

Country: ln

Cpu: c0

ACD VERSION

release : 8

bug_fixing : 1

protocol_id : 90

version_dy_hr_stat : 11

(3)pcx2a> <this is Logged in screen, system is ready to accept the instructions>

General Commands

Command “config”

Use the config command to display the peripheral cards installed in the media gateway.

- a) config <all> - shows the status of all MG
- b) config <ACT no.> - shows the status of designated MG
- c) config <ACT no.> <Slot no.> - shows the status of designated card

The cplstat Command

Information:

The cplstat command allows to check the status of the board in the rack or Crystal shelf.

Syntax:

cplstat <number of the rack or Crystal shelf> <number of the board>

Example:

Viewing the status of the PRA DASS2 board in slot N°6 of rack N°3.

```
(1)OXE0> cplstat 3 6
  Logic coupler type : CPL_DASS2 (29);
  MAO coupler type : DASS2 ;
  Coupler state : IN SERVICE ;
  ..... : .....
```

Command “rstcpl”

Syntax : rstcpl < Crystal number><board position>

The **rstcpl** command is used to reset the board and used if any abnormality is observed.

cplalarm command

Syntax : cplalarm < Crystal number><board position>

The **cplalarm** command is used to output the board alarm counters.

cplalarm command example

```
(1)pabx56> cplalarm
select pour cr 0 cpl 0
```

Compteurs d'alarme, cristal 0 coupleur 0 :

Date de debut du comptage : 06/04/00 a 20:10:57

Compteur 1 = 0	Compteur 2 = 0	Compteur 3 = 0
Compteur 4 = 0	Compteur 5 = 0	Compteur 6 = 0
Compteur 7 = 0	Compteur 8 = 0	Compteur 9 = 0
Compteur 10 = 0	Compteur 11 = 0	Compteur 12 = 0
Compteur 13 = 0	Compteur 14 = 0	Compteur 15 = 0

```
(1)pabx56>
```

Meaning of the alarm counters :

Alarm counters for a PRA type board :

- counter 1 : SIA2M
- counter 2 : MS
- counter 3 : PVT
- counter 4 : FEC
- counter 5 : TE
- counter 6 : ATD

Alarm counters for a PCM type board :

- counter 6 : AMTD
- counter 8 : PVMT
- counter 9 : SIA64K
- counter 11 : ATD
- counter 12 : TE
- counter 13 : PVT
- counter 14 : SIA2M
- counter 15 : MS

The trkstat Command

Information:

The **trkstat** command is used to display a list of all link trunks (traffic and virtual channels) with their status: **F** for free, **B** for busy.

Syntax:

trkstat[-r] <number of the trunk group>

The -r option is used to refresh data.

Example:

Viewing the status of the trunks of a DASS2 link identified by trunk group N°4.

(1)OXE0> **trkstat** 4

```

=====
=====+
| T R U N K   S T A T E   Trunk group number : 4           |
|                               Trunk group name   : dass2   |
|                               Number of Trunks    : 30      |
+-----+

```

```

| Index :   1   2   3   4   5   6   7   8   9  10  11  12  13 |
| Neqt  : 667 668 669 670 671 672 673 674 675 676 677 678 679 |
| State :   F   F   F   F   F   F   F   F   F   F   F   F   F |
+-----+
| Index :  14  15  16  17  18  19  20  21  22  23  24  25  26 |
| Neqt  : 680 681 683 684 685 686 687 688 689 690 691  692 693 |
| State :   F   F   F   F   F   F   F   F   F   F   F   F   F |
+-----+
| Index :  27  28  29  30                                     |
| Neqt  : 694 695 696 697                                     |
| State :   F   F   F   F                                     |
+-----+
| F: Free |   B: Busy | Ct: busy Comp trunk | Cl: busy Comp link |
|          |          | Cr: busy Comp trunk for RLIO inter-ACT link |
+-----+

```

Result:

In the example, the DASS2 link is represented by the No. 4 trunk group which is called DASS2_1. The DASS2 link has 30 channels of traffic which are numbered from 1 to 30. All the channels are in free mode (F).

Command incvisu

Use the incvisu command to display the main queue or outfall queue incidents (errors) saved in local mode on the management console.

By default, the incvisu command displays the incidents transmitted since the last system restart.

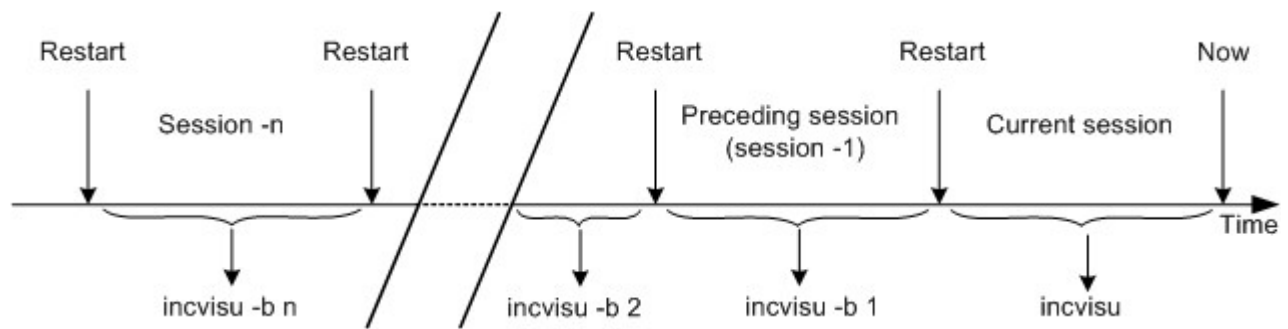
Syntax:

```

incvisu -b n Displays the incidents preceding the last "n" session restarts
incvisu -b 1 Displays the incidents of the last session
incvisu      Displays the incidents of the current session

```

Example of system restart with incvisu command



The following table gives the options that can be used with the incvisu command and their meaning.

Meaning of incvisu command optional parameters

Optional parameter	Meaning
-h	Displays the help menu
-s	Displays the incidents of the starting phase <i>Example 1: incvisu -s edits the startup incidents of the current session</i>
-o	Displays the incidents of the outfall queue <i>Example 2: incvisu -o edits the outfall queue incidents of the current session</i>
(no parameter)	Displays the incidents of the starting phase and the main queue when neither "-s" nor "-o" options are used
-b n	Displays the incidents of the n backup session <i>Examples 1:</i> - incvisu -b 1 edits the main queue incidents of the previous backup session - incvisu -b n edits the main queue incidents of the n backup session
-t n (n being a tail size)	Displays the last "n" incidents <i>Example 3:</i> incvisu -t 100 edits the last 100 starting phase and/or main queue incidents of the current session
-i x,y	Displays the incidents between markers x and y
-r	Displays incidents in reverse order, i.e. from the most recent ones to the oldest ones <i>Example 4: incvisu -r displays all the incidents in reverse order, from the most recent ones to the oldest ones</i>

Optional parameter	Meaning
-e "character string"	Displays incidents containing a specific character string <i>Example 5: incvisu -e "character string" (eg. "Out Of Service") displays only the current session incidents where such character string appears</i>

You can use command incvisu with several different optional parameters.

Examples 2:

- *incvisu -b 1 -t 100 -r -e "out of service" displays the last 100 incidents of the last session with the character string "out of service" in reverse order,*
- *incvisu -b 1 -s edits the session start file incidents of the last backup session,*
- *incvisu -t 100 -o displays only the outfall queue incidents.*

The results obtained with the command can be sent to a file, using the following command, for example: incvisu > incident.txt. The generated file can then be transferred and used as a text file.

The "grep" option may be used with the incvisu command as follows:

- *incvisu | grep US to display incidents with the character string "US",*
- *incvisu | grep 2064 to display incidents with the character string "2064".*

Example of use of the incvisu command:

```
(1)pabx56> incvisu —T 10
06/03/00 15:34:53
```

```

-----
11/03/00 08:34:29 M|0/11/0/000|=4:2523=La version de
boot de ce coupleur est incorrecte
11/03/00 08:34:29 M|0/11/0/254|=2:3747=Timeslot inconnu
11/03/00 08:34:29 M|0/11/0/254|=2:3756=Message perdu
11/03/00 08:34:33 M|0/11/0/001|=2:2151=Subdevice V24 present non declare
11/03/00 08:34:34 M|0/11/0/000|=2:2151=Subdevice MAC/PC
present non declare
11/03/00 08:34:34 M|0/11/0/001|=2:2151=Subdevice MAC/PC
present non declare
```

11/03/00 08:34:34 M|0/11/0/004|=2:2151=Subdevice MAC/PC
present non declare
11/03/00 08:34:35 M|0/11/0/001|=2:2151=Subdevice AOM40
present non declare
11/03/00 08:34:40 M|-/——-/———|=5:2141=Qte ressources SET normale
11/03/00 08:35:30 M|0/11/0/254|=5:2019=Mise en service d'un coupleur UA

“intipstat” command

This command is used to check the status of inter-ACT connections via INT-IP.

Main menu

>intipstat

Int IP informations :

Display an INTIP structure : 1
Display all INTIP structure : 2
Display INTIP MAC hash table : 3
Quit this tool : 0
Enter your choice :

Display an INTIP structure

This option is used to check an INT-IP connection designated by the ACT number and the position of the INT-IP board. The characteristics and the status of the board are displayed and also information on the inter-ACT connection(s).

Example:

Enter your choice : 1
Cristal Number : 0
Coupler Number : 12
Coupler INTIPA cr 0, cpl 12, Ghost Eqt 2121
IP Address : 192.40.8.33
Netmask : 255.255.255.0
Gateway Address : 192.40.8.254
Mac Address : 00:80:9f:04:44:ab
Sign. Ip Term. Load : 3
Sign. IRC. Load : 3
Domain IP : 0
Local IP Version : IPV4
IP QoS Category : 0
Embedded Ethernet : NO

Max Comp Allowed : 6
INTER ACT LINK :
cr opp : 2
cpl opp : 10
status int : INT_A
link_right : UP
used_ts : 0
bkUp access nb. : 255
bkUp Ts nb. : 255
volume_system : 1
local_volume : 0
INTIPA list inter act link :
INTER ACT LINK :
cr opp : 2
cpl opp : 6
status int : INT_A
link_right : UP
used_ts : 0
bkUp access nb. : 255
bkUp Ts nb. : 255
volume_system : 1
local_volume : 0
INTER ACT LINK :
cr opp : 4
cpl opp : 6
status int : INT_A
link_right : UP
used_ts : 0
bkUp access nb. : 255
bkUp Ts nb. : 255
volume_system : 1
local_volume : 0

Display all INTIP structure

This option gives the same information as the previous option but for all a PBX's INT-IP boards. This option is the same as the “intipstat list” command.

Running the ippstat Command

The **ippstat** command is used to view IP phone data.

The "ippstat d numann" command is used to view data for the IP phone with directory number "numann". The result is identical to item 2 in the following menu.

Main Menu

>ippstat

IPP (IP Phone) information :

```
-----
Display the IP Address of the local node      : 1
Display IP Phone data of a set                : 2
Display all IP Phone on local node            : 3
Display registration terminaisons(short info): 4
Display registration terminaisons (long info): 5
Display an IP Phone Board structure           : 6
Display all IP Phone Board structure          : 7
Display all Mac Address Used                  : 8
Display all Mac Address with invalid channels: 9
Display Ip phone by adr mac/adr adr ip/mcdu  : 10
Display number of IP Phones connected         : 11
Display all IP Phones connected on a board    : 12
Display all IP Phones connected with boards   : 13
Ip Phone download Menu                       : 14
Timeout for telnet session of NOE set        : 15
INT IP Menu                                 : 16
Domain Menu                                 : 17
IPlink Menu                                 : 18
Status Menu                                 : 19
Display boards encryption parameters         : 20
Enable telnet service for GD/GA              : 21
Quit this tool                               : 0
```

Enter your choice :

Displaying the IP Address of the Local Node

Item 1 is used to display the IP address of the local CPU.

Example:

Enter your choice : 1

IP Address of this node is : 191.254.1.2

Displaying the IP Address of an IP Phone

Item 2 is used to display the IP address of an IP phone.

Example:

Enter your choice : 2

Enter a directory number you want to manage : 2010

Logic coupler type : CPL_UA
MAO coupler type : UA_FICTIF
Coupler state : IN SERVICE
Terminal state : IN SERVICE
Terminal type : IP PHONE

```
+-----+ Cry:Cpl:ac:term | neqt |
nulog| typ term | dir nb | Out of service cause |
+-----+
| 19 1 0 8 | 0942 | 7| NOE_C| 13006 | ..... |
+-----+
|A: att_mserv, C: hs_defich, I: hs_isolauto, X: hs_isolman, T: hs_terdef |
|U: hs_usdef, P: hs_errparite, B: hs_bascul, Y: hs_cristisol, N: hs_inex |
+-----+
```

```
+---+-----+---+-----+-----+-----+
|Indx| Mac Address | |Neqt| QMCDU | Name | Ip Address |Type|
+---+-----+---+-----+-----+-----+
|0001|00:80:9f:56:06:a2:00|V|00270|13005 |NOE |192.168.004.055| NOE|
|Sign| 19/ 1 | |NoCnx| ON CPL | | |
+---+-----+---+-----+-----+-----+
```

Ip domain is : 0

Ip Voice Coding is IntraDom G723 : ExtraDom G723

QOS parameters for category 0:

DSCP diffserv : 0

802.1Q used No

Vlan Identifier (Vid) : 0

User priority (8021Q pri.) : 7

Firmware version is 01.09.8 on a 100 Mega hardware

Encryption : No

Firmware version Data is 03.60.30

IP Touch ram size = 8

IP Touch rom size = 8
IP Touch reset cause = 1

The first part is similar to the results obtained with the tradna or tradedq commands. However, note that the IP phone is attached to Shelf 19 Coupler 1. This is a virtual coupler declared to simplify IP phone management.

You can also find various data items concerning set domain, the compression algorithms used, and Quality of Service parameters, etc.

Displaying IP Phones Assigned to the Local Node

Item 3 is used to display a list of IP phones registered on the local node.

Displaying Operational IP Phones

Item 8 is used to display a list of operational IP phones.

Example:

Enter your choice : 8

ALL MAC ADDRESS USED

Indx	Mac Address	Neqt	QMCDU	Name	Ip Address
0001	00:80:9f:3b:7c:ce:00	V	00803 3002	IP-P	192.168.004.055
0002	00:80:9f:3b:7c:d0:00	-	00804 3003	IP p	000.000.000.000
0003	00:80:9f:04:8b:7d:00	-	00805 3004	IP-P	192.168.004.232

In the single column

V: means the channel is valid

-: means the channel is not valid

#3 mac address used

#1 with valid channel

Display Out-of-Service IP Phones

Item 9 is used to display a list of out-of-service IP phones.

Example:

Enter your choice : 9

ALL MAC ADDRESS USED WITH UNVALID CHANNEL

Indx	Mac Address	Neqt	QMCDU	Name	Ip Address
------	-------------	------	-------	------	------------

Indx	Mac Address	Neqt	QMCDU	Name	Ip Address	Type
0003	00:80:9f:56:06:a2	- 00937	13011	london jack	000.000.000.000	Ipt

In the single column

V: means the channel is valid

:- means the channel is not valid

#1 mac address used with unvalid channel

Displaying the Number of IP Phones Connected

Section 11 displays the number of sets connected to the node.

Displaying Sets Connected to the Boards

Sections 12 and 13 are used to display the sets connected to a given board or to all boards.

Enter your choice : 12

Cristal Number : 2

Coupler Number : 1

Coupler INTIPA cr 2, cpl 1, Ghost Eqt 10517

QMCDU	Name	Mac Address	Neqt	Ip Address	
3004	IP-Phone 401	00:80:9f:04:8b:7d:00	S	00805	192.168.004.232

In the single column

S: means sig. channel is supported by this board for these Ip phones

V: means voice channel is supported by this board for these Ip phones

B: means both channel is supported by this board for these Ip phones

Actually this board supports 01 signalling links

Actually this board supports 00 voice connections

Actually this board supports 00 both signalling and voice

Displaying IP Board Characteristics

Section 16 is used to display IP board characteristics.

Enter your choice : 16

Int IP informations :

```

-----
Display an INTIP structure           : 1
Display all INTIP structure         : 2
Display INTIP MAC hash table        : 3
Back to Main menu                   : 0
Enter your choice :

```

Status Menu

Section 19 Status Menu provides the status of various VoIP devices on the network (IP boards, IP phones).

The “shutdown” command

- **Overview**
This command is used to reset the CPU of the PABX that the command is performed on.
- **Command settings**
(0)A4400a>shutdown
This command resets and reboots the CPU after a fixed 30-second timeout.
(0)A4400a>shutdown —i
This command resets and reboots the CPU immediately (i stands for “immediately”).
(0)A4400a>shutdown —h
This command shuts down the CPU (h stands for “halt”). The CPU is not reset and rebooted until a key is pressed.
(0)A4400a>shutdown —p
This command simulates a power supply alarm (loss of PSAL) with a fixed delay of 1 second (this can be programmed with the “setshutdelay” command).

twin Command

```

> twin
Usage : twin [Redundancy Cpu Enable (y/n)]
Role and CPU positions:
Role of the CPU : MAIN

```

```

CPU position      : 06                      (1)
CPU address       : 132.169.4.52
Twin CPU position : 10

```

Twin CPU address : 132.169.5.54

Redundancy State:

Duplicated configuration : YES (2)

Wished sig. transfer mode : C1 signalling channel (3)

Used sig. transfer mode : C1 signalling channel

Transmission CPU-CPU : READY (4)

Telephony redundancy : READY (5)

monitel redundancy : READY (6)

memloader redundancy : READY (7)

All applications redundancy : READY (8)

The twin command displays the following information:

1. Status of the local Com Server: main or standby.

CPU position: position of the Com Server in the shelf. On Common Hardware, CPU positions are virtual.

CPU address: Com Server IP address.

2. Configuration status reminder.
3. Irrelevant.
4. status of the inter-Com Server link: **READY** or **KO**.
5. Status of the telephone application: **READY** or not ready (**KO**) for switchover.
6. Information reserved for Alcatel-Lucent support.
7. Information reserved for Alcatel-Lucent support.
8. Information reserved for Alcatel-Lucent support.

role Command

This command is used to obtain the status (main/standby) of the Com Server on which the command is run.

The command “role -b” is used to know whether the standby Com Server is standby. The twin command gives a more accurate summary of duplication status.

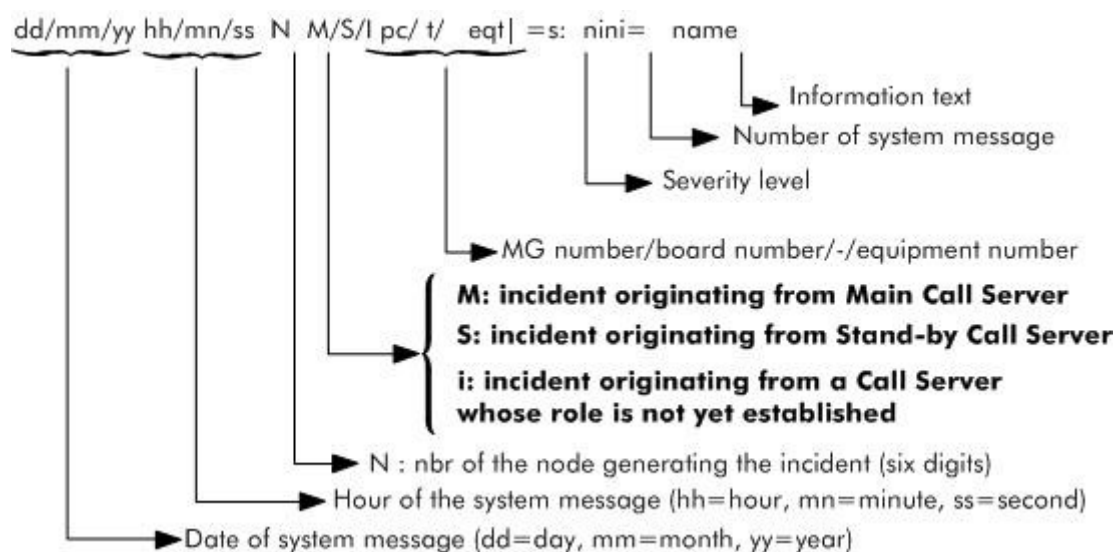
Incidents on the Standby Com Server

When a fault is detected on a standby Com Server, it sends a main Com Server incident message that is displayed with the letter “S”, this enables the user to differentiate such incidents from those originating on the main Com Server “M”.

When a standby Com Server cannot process the MAO requests transmitted by a main Com Server (this is the case when the databases are inconsistent), it signals this situation by transmitting an incident to the main Com Server.

Incident messages are displayed as follows:

Com Server Incident Messages



When Com Server role is “UNDEFINED”, the incidents transmitted are stored on this Com Server. When it switches to standby status, the incidents are sent to the main Com Server.

Command “pcsview”

pcsview	display information on the Passive Communication Server(s):
	• name • IP address

	• state • reset type • attached domains • secured domains
--	--

pcsview

PCS Name	Address	State	Reset Type
PcsNancy	10.3.16.1	INACTIVE	Default (30s)
PcsMetz	10.4.16.1	ACTIVE	Default (30s)
PcsSiltzheim	10.5.16.1	UNDEF	Default (30s)

Possible states of a PCS are:

- **INACTIVE**: the signaling link between the PCS and the Call Server is up and the PCS is in inactive state.
- **ACTIVE**: the signaling link between the PCS and the Call Server is down and the PCS should be in active state.
- **INACTIVE ***: the signaling link between the PCS and the Call Server is reestablished but the PCS is still in active state (the PCS will reboot at expiration time).
- **UNDEF**: the signaling link between the PCS and the Call Server has never been established. The PCS is not in the capacity to switch to active state.

CONCLUSION

This study projects gives the opportunity to understand the installation, design and working of SERVER BASED ALCATEL LUCENT PBX. The system also gives the scope to study the convergence of voice and data technologies through VoIP protocol. The central server maintains and controls the Remote location units and RLU's are also equipped with Passive Communication Servers in case of main call server failure. The server also integrates different application software like Call billing software, Complaint handling system and inbuilt voice mail facility. The project also includes the various maintenance commands and response of server and media gateways for the same. However the system can optimized and can be used to full utilization if it integrated in the existing data network of BSL which provides the true meaning of convergence and IP phones and conference systems can be installed and logged in every end point.